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ARTICLE

Virus study for continuous low pH viral inactivation inside a coiled flow inverter

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Funding information

German federal ministry of education and research, Grant/Award Number: project WiPro, BMBF, 031A616M; European Union (European Regional Development Fund); The German Federal State North RhineWestphalia (NRW), Grant/Award Number: NRWZiel2, 005-1010-0009

Abstract

Continuous processing for the production of monoclonal antibodies (mAb) gains more and more importance. Several solutions exist for all the necessary production steps, leading to the possibility to build fully continuous processes. Low pH viral inactivation is a part of the standard platform process for mAb production. Consequently, Klutz et al. introduced the coiled flow inverter (CFI) as a tool for continuous low pH viral inactivation. Besides theoretical calculations of viral reduction, no viral clearance study has been presented so far. In addition, the validation of continuous viral clearance is often neglected in the already existing studies for continuous processing. This study shows in detail the development and execution of a virus study for continuous low pH viral inactivation inside a CFI. The concept presented is also valid for adaptation to other continuous viral clearance steps. The development of this concept includes the technical rationale for an experimental setup, a valid spiking procedure, and finally a sampling method. The experimental results shown represent a viral study using xenotropic murine leukemia virus as a model virus. Two different protein A (ProtA) chromatography setups with varying pH levels were tested. In addition, one of these setups was tested against a batch experiment utilizing the same process material. The results show that sufficient low pH viral inactivation (decadic logarithm reduction value >4) was achieved in all experiments. Complete viral inactivation took place within the first 14.5 min for both continuous studies and the batch study, hence showing similar results. This study therefore represents a successful virus study concept and experiment for a continuous viral inactivation step. Moreover, it was shown that the transfer from batch results to the continuous process is possible. This is accomplished by the narrow residence time distribution of the CFI, showing how close the setup approaches the ideal plug flow and with that batch operation.

Abbreviations: CFD, computational fluid dynamics; CFI, coiled flow inverter; CHO, chinese hamster ovary; CIP, cleaning in place; CoG, cost of goods; Cond, conductivity; CSTR, continuously stirred tank reactor; CV, column volume; HL, homogenization loop; ID, inner diameter; KCl, potassium chloride; LRV, decadic logarithm reduction value; LVP, large volume plating; mAb, monoclonal antibodies; MTP, multititer plates; Mv1Lu, mink epithelial lung cells; NaOH, sodium hydroxide; ProtA, protein A chromatography; RT, residence time; RTD, residence time distribution; SH, straight helix; ST, straight tube; TCID₅₀, 50% tissue culture infectious dose; UO, unit operation; xMuLV, xenotropic murine leukemia virus.

Symbols: d , decadic logarithm of the serial dilution steps; D , sample predilution; n , total number of analyzed cell culture wells; n_p , number of virus-positive wells; p , probability value; P_i , summation of virus-positive cell cultures within the virus transition area; R_w , relative width; v , decadic logarithm of the volume conversion factor; V , overall volume of the virus-containing sample; V_w , analyzed sample volume; Y_0 , decadic logarithm of the highest dilution of the virus-containing sample; $\Theta_{0.005}$, dimensionless time point where 0.5% of the maximum dimensionless concentration is reached; $\Theta_{0.995}$, dimensionless time point where 99.5% of the maximum dimensionless concentration is reached.

KEYWORDS

coiled flow inverter, continuous processing, low pH, monoclonal antibodies, viral inactivation

1 | INTRODUCTION

In the last few decades, the continuous production of monoclonal antibodies (mAb) has been one of the most promising developments in the industry (Hernandez, 2015). In combination with single-use equipment, modern and flexible plants with a shorter time-to-market, improved cost of goods (CoG), and lower susceptibility toward cross contaminations can be realized. The possibility to build a fully continuous mAb platform process on the miniplant scale was shown in the MoBiDiK project (Klutznick, Magnus et al., 2015). Moreover, the CoG benefits of a continuously operated downstream have been discussed (Klutznick, Holtmann, Lobedann, & Schembecker, 2016). Although most unit operations (UOs) are available for continuous processes and in disposable equipment, the continuous viral clearance is often not taken into account (Challener, 2014).

Effective viral clearance is mandatory for all biopharmaceutical production processes to guarantee patients' safety. Therefore, master and working cell banks as well as all process feed streams are controlled thoroughly (ICH Expert Working Group, 1999). Nevertheless, the production process itself has to be able to reduce the possible amount of viruses. For the production of mAb at clinical and production scales, at least two orthogonal dedicated viral clearance steps are necessary (EMA, CHMP, & BWP, 2009; FDA, 1997; ICH Expert Working Group, 1999). Each step has to reach a decadic logarithm reduction value (LRV) ≥ 4 (EMA, 1996). Viral clearance steps can be divided into different categories, depending on their target and UO used. The process has to be capable of removing different types of viruses. Lipid-enveloped viruses are sensitive towards physicochemical treatments, like extreme pH and solvent/detergent, that affect the integrity of the envelope membrane, and the viral particles become non-infectious. Small non-enveloped viruses do not respond to these treatments and often have to be removed through viral filtration or chromatography. The UOs used for viral clearance can be divided into two groups: fortuitous techniques combine product purification and viral clearance properties, whereas deliberate techniques have solely been introduced into the process for their viral clearance capabilities (Farshid, Taffs, Scott, Asher, & Brorson, 2005). Chromatography steps, such as protein A (ProtA) and anion exchange chromatography, as well as precipitation are fortuitous techniques (Brorson, Brown, Hamilton, & Stein, 2003a; Curtis, Lee, Blank, Brorson, & Xu, 2003; Farshid et al., 2005). Deliberate techniques include heat, solvent/detergent treatment, virus filtration, low/high pH, chemicals, and UV-C irradiation (Farshid et al., 2005).

Low pH viral inactivation leads to the inactivation of enveloped viruses through capsid and membrane structure disintegration (Brorson et al., 2003b). It plays an important role

within mAb production processes, as most mAb are stable at pH conditions ≤ 4 (CMC Biotech Working Group, 2009). Moreover, the eluate stream from ProtA chromatography, the first purification step within the mAb production process, usually has a pH ≤ 4 without the necessity of further adaptation (Kelley, Blank, Lee, 2009). Therefore, low pH is one of the two platform process viral clearance steps. Furthermore, it is one of the most important viral clearance steps addressing the enveloped murine leukemia virus, the relevant model virus for retrovirus-like particles found in the production cell lines like Chinese hamster ovary (CHO) cells (CMC Biotech Working Group, 2009).

So far, the viral clearance validation of continuously operated downstream process steps has been discussed (Johnson, Brown, Lute, & Brorson, 2017) but not investigated in detail. Already, existing batch techniques have to be adapted to the continuous environment. Therefore, this study presents a first attempt to develop a valid technique for virus studies for continuous viral clearance steps. With the help of a coiled flow inverter (CFI), the low pH inactivation for the mAb platform process was realized for continuous processing. The detailed experimental setup and the results are shown in the following sections.

2 | MATERIALS AND METHODS

In this section, all used materials and the corresponding methods are described.

2.1 | Materials

2.1.1 | Feed and buffers

All chemicals for buffer preparation were provided by Sigma-Aldrich (St. Louis, MO), Fischer Scientific (Schwerte, Germany), and Merck Millipore (Billerica, MA). Buffer solutions were stored using Sartorius Stedim (Göttingen, Germany) Flexboy bags. The feed was derived by continuous fermentation in the framework of the MoBiDiK project and contained 0.72 to 0.74 g/L mAb. All buffers and the feed were used after 0.2 μ m filtration.

2.1.2 | Tubings

All tubings were provided by Saint Gobain Performance Plastics (Courbevoie, France). The tubings were C-Flex, Sanipure BDF, and Pharmed BPT biopharmaceutical tubing. Connections between tubings were either derived by using I-, Y-, and reducing connectors, or by welding with a Sartorius Stedim Biotech Biowelder TC.

2.1.3 | Spike material

Spike and mock spike material was provided by Charles River Laboratories Germany GmbH (Cologne, Germany). The enveloped, xenotropic murine leukemia virus (xMuLV; strain: TEK-1) was propagated using mink epithelial lung cells (Mv1Lu [NBL-7]; ATCC CCL[®]-64[™] ATCC. Mv 1 Lu (NBL-7) (ATCCR CCL-64[™])). For this purpose, the adherent Mv1Lu cells were seeded in 175-cm² cell culture flasks in Dulbecco's Modified Eagle's Medium/F12 and directly inoculated with xMuLV virus stock solution. The infected cell cultures were incubated for 96 h at 37°C and 5% CO₂ and afterwards split into fresh cell culture flasks for another amplification cycle. The virus particle containing cell culture supernatant was removed from the cell culture flasks and transferred to falcon tubes. The supernatant was centrifuged at room temperature for 10 min at 800g. The centrifuged supernatants were aspirated, pooled, and subsequently processed via a filter with a pore size of 0.45 µm into sterile falcon tubes. These aliquots were stored at temperatures ≤−60°C until usage. Before first use, the initial titer of each xMuLV batch was determined by endpoint titration (also referred to as endpoint dilution) and determination of the tissue culture infectious dose 50 (TCID₅₀). The mock spike samples were generated in an analogous way by withdrawing the cell culture supernatant from noninfected Mv1Lu cell cultures.

2.2 | Analytics

The analytics of all experiments included pH and conductivity measurement as well as assays for virus concentration determination.

2.2.1 | pH measurements

For pH measurements, Tophit CPS471D pH electrodes from Endress +Hauser (Reinach BL, Switzerland) were used. The electrodes were calibrated prior to every experiment.

2.2.2 | Conductivity measurements

Conductivity was measured with the help of SciCon single-use luer conductivity sensors (SciLog, Madison, WM). The sensors were connected to the chromatography unit for data logging. All sensors were pre-calibrated by the use of 0.01 mol/L KCl solution.

2.2.3 | Titration and large volume plating

The quantification of virus particles was conducted by endpoint titration (also referred to as endpoint dilution) in combination with large volume plating (LVP) and determination of the TCID₅₀. For this reason, process samples were collected and pre-diluted with the respective PG-4 cell culture medium. Only non-toxic and non-interfering final dilutions (for details see Section 2.4.2) were used for the cell culture analysis and titer determination. The pre-diluted sample was further serially three-fold diluted in eight replicates using 96-well multititer plates (MTPs).

Subsequently, 100 µl of these serial dilutions were transferred onto 96-wells cell culture plates seeded with PG-4 cells. For the LVP, 200 µl of the prediluted sample was directly applied onto 96-well PG-4 cell culture plates without any serial dilution, resulting in a lower detection limit and, therefore, in a more sensitive quantification procedure. These prepared cell cultures were incubated at 37°C and 5% CO₂ for at least 5 days. Afterwards, the cells in each well of the 96-well MTP were analyzed via light microscopy to determine the virus-induced changes in cell growth and morphology (cytopathic effects).

The titer was calculated using different statistical methods depending on the number of virus-positive cultures per cell MTP detected (ICH Expert Working Group, 1999).

For samples with a high viral titer (e.g., virus stock solution or load sample) result, even with high pre-dilutions and the additional serial three-fold dilutions in virus-positive endpoint titrations (referred to as Mode A), the TCID₅₀ was calculated according to the Spearman and Kaerber formula:

$$\log_{10} \text{TCID}_{50}/\text{ml} = -\left(-Y_0 + \frac{d}{2} - d \cdot \sum P_i - v\right),$$

Y_0 represents the decadic logarithm of the highest dilution of the virus-containing sample, which still causes the simultaneous infection of all eight replicate cultures. d represents the decadic logarithm of the serial dilution steps. Therefore, d equals the value of $\log_{10} 3$. The term P_i mathematically describes the summation of virus-positive cell cultures within the virus transition area, which is the dilution range where only single cell cultures out of the eight replicates are virus-positive, starting with Y_0 . The variable v represents the decadic logarithm of the volume conversion factor, which is related to the volume per cell culture well.

Samples that were subjected to low pH treatment and withdrawn from the CFI represented process fractions with a diminished viral titer due to the inactivation of the enveloped MuLV under acidic pH conditions. Therefore, various kinetic samples were analyzed via LVP, which is characterized by a higher sensitivity. Depending on the number of virus-positive cell culture wells, two additional calculation modes were applied for the titer determination. In case less than 15% of all cell culture wells were virus-positive, the titer calculation was based on a function representing a Taylor series (mode D):

$$\text{Titer}/\text{ml} = \frac{D}{V_w} \cdot \left(-\ln\left(\frac{n - n_p}{n}\right)\right).$$

The calculation according to mode D generates a ratio between the sample predilution D and the analyzed sample volume V_w as well as between the number of virus-positive wells n_p and the total number of analyzed cell culture wells n . The multiplication of the determined ratios results in the virus titer per milliliter.

In case that no virus-positive wells were detected during microscopic analysis, the quantification of the viral titer had to be correlated to the detection limit (mode E). Consequently, in case no virus was detected, the titer per milliliter must be lower or equal (≤) than the detection limit. The calculation of the detection limit was conducted using a function related to a Poisson distribution with a

probability value of $p < 0.05$ (95% confidence limits; ICH Expert Working Group, 1999).

$$\text{Titer/ml} \leq \frac{\ln p}{V \cdot \ln\left(1 - \frac{V_w}{V}\right)}$$

The variable V_w represents the volume analyzed, which is directly correlated to the volume of the pre-diluted process sample per 96-well culture analyzed in the LVP assay multiplied with the number of wells and the pre-dilution factor of the virus-containing sample. The variable V represents the overall volume of the virus-containing sample, in this respective case the amount of liquid withdrawn from the CFI.

To calculate the LRV within one sample, the determined virus titer for the respective sample was subtracted from the virus titer of the corresponding load sample.

2.3 | Plant setup

The overall plant setup can be seen in Figure 1. It consists of a laboratory-scale BioSMB unit for continuous ProtA chromatography, a homogenization loop (HL), and finally the CFI. Several pH and conductivity sensors are used to monitor the process.

At first, multiple ProtA chromatography columns are operated simultaneously, providing a continuous outlet elution stream from

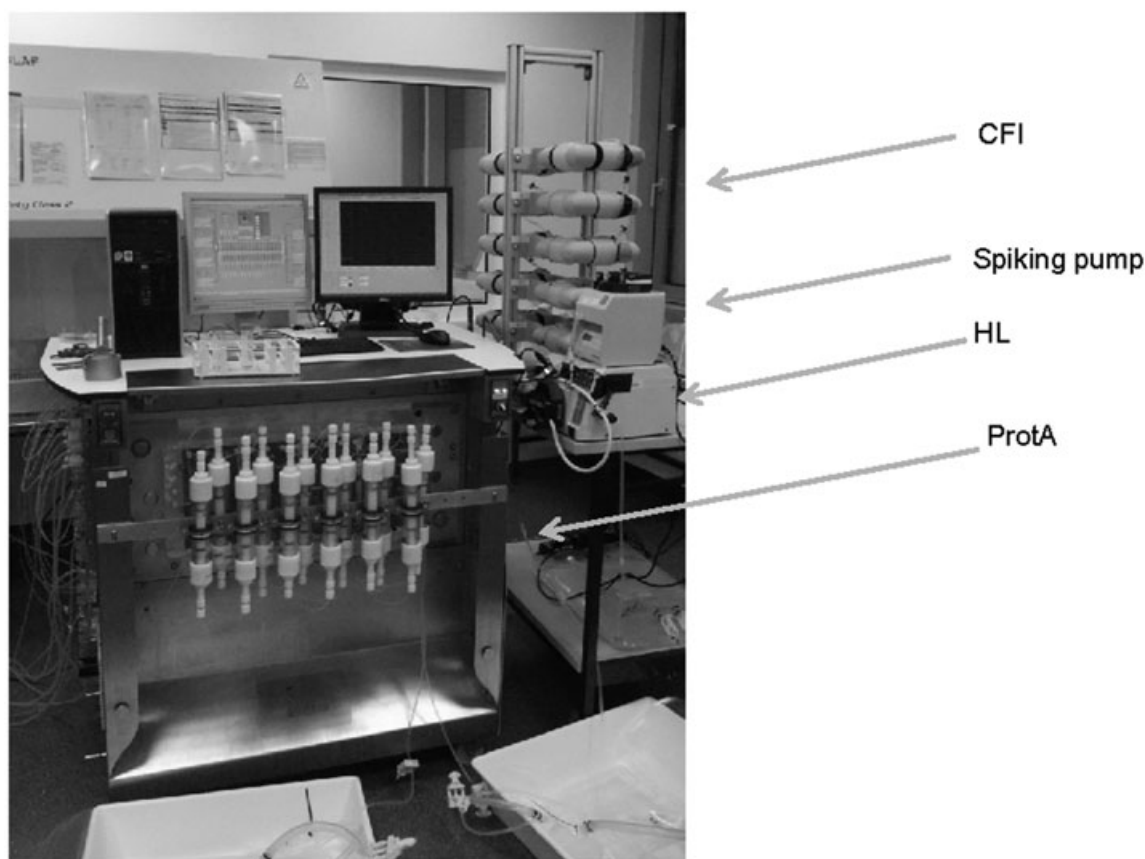
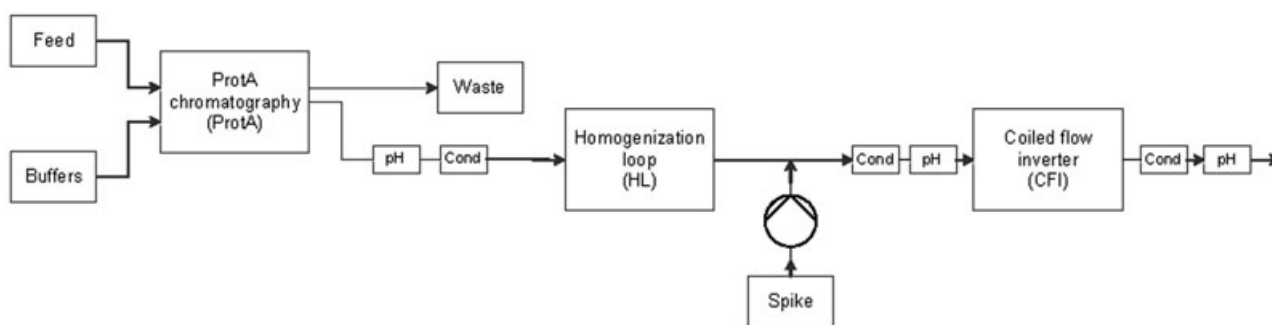


FIGURE 1 Process scheme and experimental setup for the viral study. CFI: coiled flow inverter; cond: conductivity sensor; HL: homogenization loop; pH: pH sensor; ProtA: protein A

the BioSMB unit. Each column passes the common batch step sequences including equilibration, load, wash, elution, regeneration, and cleaning in place (CIP). As product peak cutting is not conducted for continuous chromatography, the elution profile in the product outlet shows a periodically fluctuating pH level between the elution buffer (pH 3.1) and the wash 2 buffer (pH 7.0).

In accordance with the mAb production process, the low pH viral inactivation step is placed after the ProtA chromatography unit. As the overall pH for low pH viral inactivation should be below 4, a HL is realized for dampening the pH profile. The HL can be understood as a continuously stirred tank reactor (CSTR), leading to pH levels constantly below 4. After the HL, low pH viral inactivation takes place inside the CFI. The CFI consists of a tubing coiled around a cylindrical superstructure, leading to the formation of a secondary flow pattern called Dean vortices. After a certain number of turns, 90° bends are introduced, forcing the rebuilding of the Dean vortices. This structure leads to an enhanced radial mixing and a narrow residence time distribution (RTD). By using the CFI for low pH viral inactivation, a defined minimum residence time (RT) can be guaranteed, whereas the maximum RT is minimized, preventing product damage (Klutz, Kurt, Lobedann, & Kockmann, 2015; Klutz, Lobedann, Bramsiepe, & Schembecker, 2016).

2.3.1 | Continuous ProtA chromatography

A prototype of a Tarpon Biosystems (Worcester, MA) (now Pall) BioSMB chromatography unit was used to realize continuous ProtA chromatography. For the experiments, the BioSMB was operated with 12 columns (inner diameter [ID] 1.6 cm), each filled with 16 ml MabSelect Sure ProtA resin (GE Healthcare, Upsalla, Sweden). Membrane contactors from Liqui Cel Membrana (Model MicroModule®) were used for degassing the buffers and the cell free harvest upstream to the chromatography.

2.3.2 | Homogenization loop

The HL was made of Pharmed BPT biopharmaceutical tubing size LS17 (ID 6.4 mm). The length was set to 105.7 cm, generating a hold up volume of 34 ml. The homogenization was guaranteed by using a Masterflex L/S pump (Cole-Parmer GmbH, Wertheim, Germany) set to 100 rpm.

2.3.3 | Coiled flow inverter

The CFI was constructed to deliver a minimum RT of 120 min, meaning that not a single fluid element left the CFI before the minimum RT was reached. Therefore, the average RT was higher than 120 min and was determined by RTD characterization (see Sections 2.4.1 and 3.1).

The CFI was constructed with SaniPure BDF biopharmaceutical tubing (ID 3.2 mm). The square-shaped coil superstructure was made of polypropylene frames (outer diameter 63.0 mm, curvature ratio 21.7), allowing a maximum of 12 turns per coil. Ten of these frames

were used, creating 39 bends with a 90° bend angle. Through the use of 10 turns per coil, a length of approximately 88 m was achieved. A picture of the final CFI setup can be seen in Figure 4.

2.4 | Experimental procedure

In this section, the experimental procedure for the virus study and the necessary pre-studies (RTD characterization and pre-experimental assays) is described. The virus studies were conducted at Charles River Laboratories Germany GmbH (Cologne, Germany). The studies were performed at room temperature between 18.0 and 19.5°C.

2.4.1 | RTD characterization

In order to characterize the two UOs HL and CFI, RTD studies were performed. For these studies, step functions with MilliQ water and 1 mol/L NaCl solution were used. The step answer in conductivity was evaluated. The experimental setup can be seen in Figure 2. The flow rate was set to 4.8 ml/min by using a Masterflex pump. The step functions were performed with the help of a three-way stopcock. Three different setups were investigated: (a) the HL, (b) the CFI, and (c) the combination of HL and CFI. The conductivity step changes were executed in both possible directions. Therefore, after the complete filling with 1 mol/L NaCl solution, the experiment was performed in the other direction by filling the setups with MilliQ water at constant flux of 4.8 ml/min. All experiments were performed at least three times.

The concentration measurements were normalized and evaluated between 0.5% and 99.5% of the maximum value. The average RT was determined by numerical integration using the trapezoidal rule. The RT was then normalized with the help of the average RT. The relative width R_W was calculated according to the following equation:

$$R_W = \frac{\theta_{0.005}}{\theta_{0.995}},$$

$\theta_{0.005}$ and $\theta_{0.995}$ hereby represent the dimensionless time points where 0.5% and 99.5% of the maximum dimensionless concentration is reached. The relative width therefore shows the narrowness of the RTD and expresses how close the setup approaches ideal plug flow and with that batch operation.

2.4.2 | Pre-experimental cytotoxicity and interference assays

Prior to the experiments, cytotoxicity and interference assays were performed to determine the necessary sample dilution rate. To assess the potential influence of the test item on the growth and morphology of the detection cell line used, representative samples of one ProtA eluate pool were diluted and applied onto the cells. For this purpose, cells of the feline xMuLV indicator cell line (Bassin, Tuttle, & Fischinger, 1971; Rosenberg and Jolicoeur, 1997) PG-4 (PG-4 S+L-; ATCC® CRL-2032™ ATCC. PG-4 (S+L-) (ATCCR CRL-2032™)) were seeded into 96-well

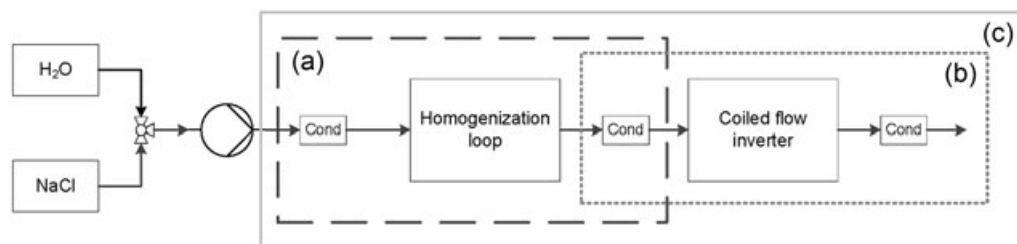


FIGURE 2 Experimental setup for RTD measurements. Setup (a) for RTD measurements of HL, setup (b) for RTD measurements of CFI, and setup (c) for combined RTD measurements. CFI: coiled flow inverter; Cond: conductivity sensor; HL: homogenization loop; RTD: residence time distribution

MTPs and cultured at 37°C and 5% CO₂. On the next day, eight replicates of serial three-fold dilutions of the test items were prepared using the corresponding cell culture medium McCoy's No5. Afterwards, the dilutions were directly transferred to the MTPs containing PG-4 cells followed by another incubation period. The in this way treated cell cultures were analyzed via light microscopy after the negative control (medium only) reached confluence. The used dilutions were assessed regarding their potential to cause cytotoxicity, to influence the cell growth, and to alter the cell morphology (cytotoxicity testing).

For assessing the potential influence of the test item on the replication of xMuLV, a non-cytotoxic dilution was used to perform the interference assay.

For this purpose, PG-4 cells were seeded into MTPs and cultured at 37°C and 5% CO₂. On the next day, xMuLV was serially three-fold diluted in cell culture medium (control medium) in eight individual replicates. Simultaneously, xMuLV was diluted in the same way in the cell culture medium containing the test item at the non-cytotoxic dilution (test media). The virus pre-dilutions were transferred to separate MTPs containing PG-4 cells and were incubated for at least 5 days. After this incubation period, the cells in each well of the MTPs were analyzed for the occurrence of cytopathic effects via light microscopy and the corresponding virus titer was calculated. The used test item dilution was evaluated as non-interfering when the difference in the virus titer was $\leq 0.5 \log_{10}$ or within the range of the 95% confidence limit compared with the titer determined with the control medium.

2.4.3 | Virus study for the continuous process

The process scheme of the virus study for the continuous process was already introduced in Figure 1. In the following sections, a more detailed view into the experimental procedure is given.

ProtA methods

Two different ProtA methods were used to investigate two different average pH levels for low pH viral inactivation. The BioSMB chronograms for both methods can be seen in Figure 3. The elution duration for both methods was set to 5 column volumes (CV) with a flow rate of 4.65 ml/min and a switch time of 17.3 min. The elution buffer with a pH value of 3.1 consisted of 100 mmol/L NaAc and 50 mmol/L NaCl. The Mode A chronogram in Figure 3 shows that elution and peak start at the same

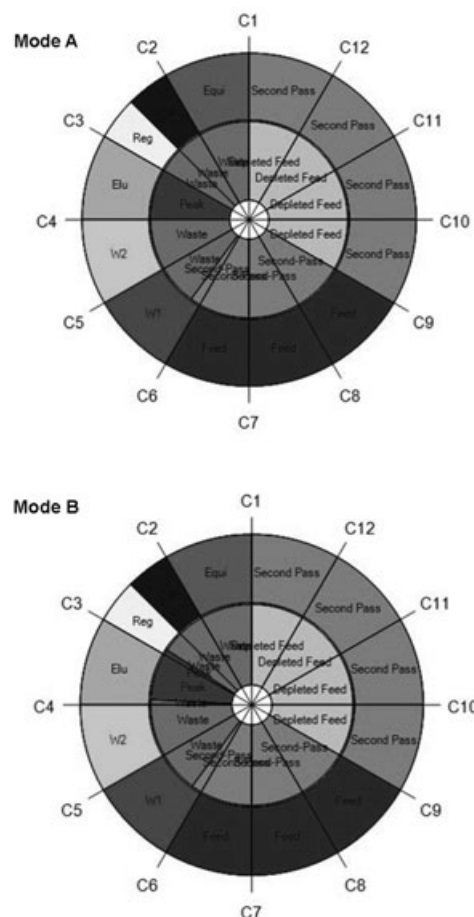


FIGURE 3 BioSMB chronograms for chromatography Modes A and B; Outer ring defines column inlet, inner ring defines column outlet; C1-C12 describes the used 12 columns. Depleted feed: flow-through of second pass which is discarded; Elu: elution; Equi: equilibration; Feed: fermenter harvest; Peak: collected product; Reg: regeneration; Second pass: flow-through of feed; W1: wash 1; W2: wash 2; Waste: flow-through of cleaning procedures

time. As the eluting column was flushed with wash 2 buffer before (50 mmol/L Tris, 50 mmol/L NaCl, pH 7.0), 1 CV of this buffer enters the product line and is then mixed with 4 CV elution buffer until the next column was switched into the elution step. Mode B reduces the amount of wash 2 buffer entering the product line by transferring the first 0.5 CV

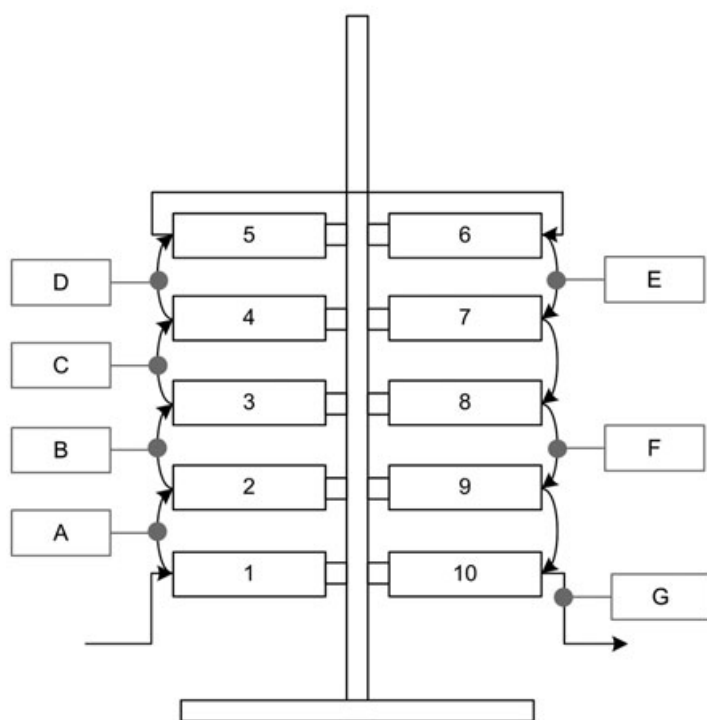


FIGURE 4 CFI setup with sampling points realized with three-way stopcocks inside the connections between two frames. CFI: coiled flow inverter

of the elution step into the waste line (lower chronogram of Figure 3). As the first 0.5 CV of the regeneration step is collected as product, 4.5 CV elution buffer are mixed with 0.5 CV of wash 2 buffer. Due to the high pH in wash 2 buffer, the overall pH level of the product stream of Mode A is higher than of Mode B.

Starting and CIP procedure

Before and after each experiment, CIP with NaOH solution took place for the CFI. After the initial CIP, the CFI was flushed with an equilibration buffer, and all air bubbles were removed. The ProtA system needed 2.5 h of initial run to reach the steady state. After the first 2.5 h of ProtA run was performed, the CFI was filled completely with spiked ProtA eluate for 3 h to reach the steady state, leading to a run-in phase of 5.5 h. The necessary duration of 3 h was determined from the corresponding RTD experiments.

Spiking procedure

The batch spiking procedure had to be realized for a continuous process. To achieve the standard spiking level of 5% (v/v), 230 μ l/min spiking material had to be added to the 4.65 ml/min ProtA stream. A Gilson Minipuls 3 pump (Gilson, Inc., Middleton, WI) was used to add the spike stream to the ProtA stream via a Y-junction. No internal mixing unit was used. The spiking point was set behind the HL for different reasons. On the one hand, the CFI

is the actual low pH viral inactivation device. Therefore, the spiking in front of the HL would have caused a potential loss in maximum LRV, as a certain RT was realized at low pH levels before the CFI was entered. Calculations before the experiments lead to an LRV estimation of about 2.6 for the HL (based on CMC Biotech Working Group (2009), results not shown). On the other hand, a decreased internal mixing at the spiking point was supposed to lead to a decreased log reduction capacity, as a poor mixing of ProtA stream and spiking stream could lead to the formation of temporally protective zones. Thus the experimental setup chosen represents the worst case for LRV determination.

Sampling method

Due to the continuous flow inside the CFI, different time points for sampling are linked to a certain location. To withdraw the samples from different time points inside the CFI, three-way stopcocks were introduced between certain frames. The sampling points as well as the final CFI can be seen in Figure 4. The time points, which are represented by the different sampling points, were determined by measuring the average RT inside the complete CFI (see Section 3.1).

As low pH viral inactivation is highly time-dependent, the reaction had to be stopped during the sampling procedure. Therefore, each sampling container was filled with 4.13 ml of

2 mol/L Tris buffer before sampling. With the help of this method, each drop of sample was neutralized immediately when entering the stirred sampling container. This prevented post-sampling viral inactivation.

Sampling inside the CFI leads to a disturbance of the internal flow patterns due to stopping of the flow. Therefore, the sampling order had to be chosen carefully to guarantee valid results. This can be achieved by using back-to-front sampling, starting with the last sampling point (G, see Figure 4) and ending with the first sampling point (A, see Figure 4). During the sampling, no fluid is further processed inside the CFI downstream of the sampling point. Therefore, a front-to-end sampling would lead to falsified results, as the flow further downstream would stop completely each time a sample is taken. Therefore, the required part of the CFI would have to be flushed and reach the steady state initially after every sampling, leading to widely increased experiment durations.

To exclude the influences of the fluctuating pH, the sampling duration was set to one switch time (17.3 min). This leads to the complete collection of all pH levels present. As the HL serves as a CSTR, no concrete identification of the sampled column inside the CFI was possible. To exclude the influence of different ProtA columns, some sampling points inside the CFI were double-sampled. The first sample was taken after one switch time (17.3 min). Afterwards, the sampling was paused for half a switch time (8.65 min), before the sampling was continued at the same port for another switch time. During the pausing of sampling, the three-way stopcock position was not changed. The continuing flow through the sampling point was fed into a waste container. With this procedure, no residual fluid was trapped inside the sampling system. Residual fluid could be exposed to low pH levels for a prolonged RT and therefore falsify the results.

2.4.4 | Virus study batch process

To compare the performance of the CFI with the standard batch low pH inactivation procedure, a batch sample was drawn from the continuous ProtA chromatography unit. Therefore, the second chromatography mode (Mode B) was used, where 4.5 CV of eluate was mixed with 0.5 CV of wash 2 buffer. The fraction from one single column was taken for one switch time (17.3 min) and the pH was not neutralized. The resulting volume of 84.34 ml with pH 3.55 was then transferred to the batch experiment. After spiking a 50 ml sample with virus stock, samples were taken at the same discrete time points as in the CFI. The samples were analyzed by endpoint titration as well as LVP.

3 | RESULTS AND DISCUSSION

In the following, the results of all experiments performed are shown and discussed.

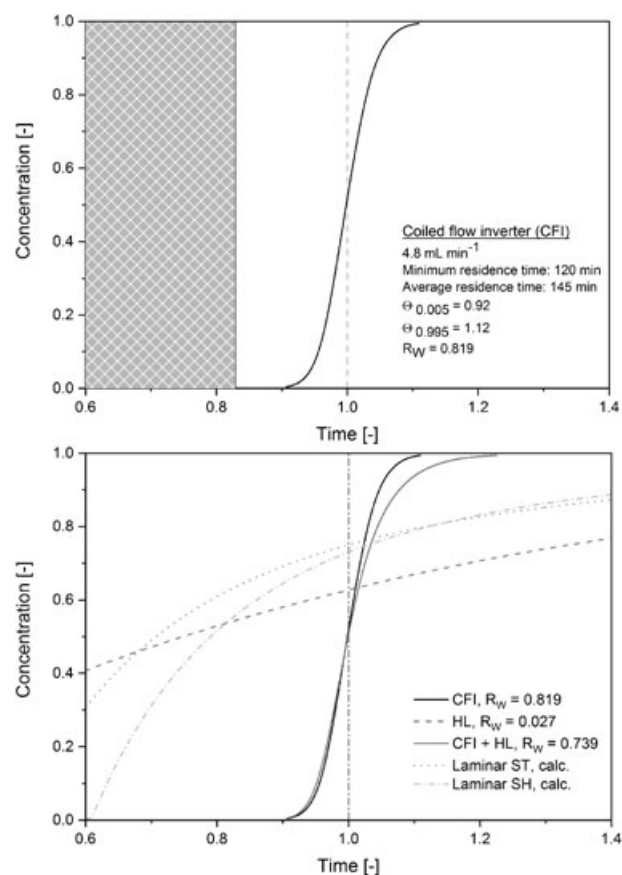


FIGURE 5 Residence time distribution characterization; upper diagram: RTD curve of CFI with minimum RT of 120 min (grey area shows time points < 120 min); lower diagram: RTD comparison of experimental results (CFI, HL, and the combination of both) with calculated RTD curves (laminar straight tube [ST] and laminar straight helix [SH]). CFI: coiled flow inverter; HL: homogenization loop; RT: residence time; RTD: residence time distribution

3.1 | RTD characterization

RTD studies with water and NaCl solution were performed to characterize the HL and the CFI. With the help of these results, the RTs corresponding to the sampling points were determined. The results of the RTD measurements can be seen in Figure 5. The CFI was designed to deliver a minimum RT of 120 min. The upper diagram shows the RTD characterization results for the CFI. The average RT was determined to be 145 min with a relative width R_W of 0.819. Consequently, every frame out of the 10 frames realized inside the CFI provides a RT of 14.5 min. The sampling positions chosen and the corresponding RTs are shown in Figure 4 and Table 1, respectively. The RTD inside the CFI is very narrow, leading to a minimized maximum RT of every single fluid element (small tailing).

TABLE 1 Sampling positions and corresponding residence times

Sample point	A	B	C	D	E	F	G
Residence time at the sampling point (min)	14.5	29	43.5	58	87	116	145

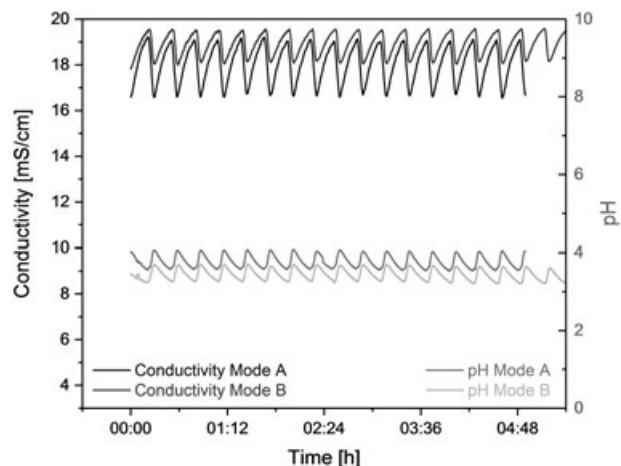


FIGURE 6 pH and conductivity experimental data measured at the entrance of the CFI (modified due to different measurement systems). CFI: coiled flow inverter

The lower diagram of Figure 5 shows the comparison of the experimental results of CFI, HL, and the combination of both with calculated RTD curves for laminar straight tube (ST) and laminar straight helix (SH). The HL shows the typical CSTR RTD behavior. The average RT is 8.12 min and the R_W is 0.027. The gray curve in the lower diagram shows the measurement of the HL in combination with the CFI. In comparison with the CFI results shown in black, the average RT increases to 153 min and the relative width decreases to 0.739. The combination of HL and CFI leads to a visibly wider RTD than using the CFI alone. Nevertheless, both constellations show a significantly better RTD behavior than the alternatives, the laminar ST and the laminar SH.

3.2 | Pre-experimental cytotoxicity and interference assays

Before the experiments, cytotoxicity and interference assays were performed. The necessary final dilution of the sample was determined to 1:54. At this dilution factor, no effect on cell growth and virus replication could be observed.

3.3 | Virus study continuous process

At first, the continuous experiments with two different ProtA modes will be discussed. The pH and conductivity measurements can be seen in Figure 6. The pH and conductivity measurements of both ProtA modes show the periodically fluctuating behavior, as described in Section 2.3. Moreover, the dampening effect of the HL can be seen clearly (initial pH fluctuation in front of the HL between 3.1 and 7). The pH values and conductivity at the exit of the CFI are not shown in Figure 6. After starting the sampling procedure, the liquid stays inside the pH measurement probe at the exit for the rest of the experiment, leading to non-representative constant pH and conductivity measurements.

TABLE 2 pH and conductivity values for Mode A and Mode B chromatography measured at the entrance of the CFI

Protein	pH measured at CFI entrance			Conductivity measured at CFI entrance (mS/cm)		
	Min	Max	Average	Min	Max	Average
Mode A	3.53	4.07	3.77	16.54	19.22	18.01
Mode B	3.20	3.71	3.44	17.98	19.62	18.91

Note. CFI: coiled flow inverter.

The differences in minimum, maximum, and average in both pH and conductivity values of Mode A and Mode B at the entrance of the CFI can be seen in Table 2. The average pH of Mode A is higher by 0.33 than in Mode B, whereas the average conductivity is lower by 0.90 mS/cm in comparison with Mode B. The virus titer and LRV results of the two experiments can be seen in Figure 7. The upper part of the diagram shows the results of the single sampling with Mode A of ProtA chromatography. The first two columns represent the virus titer results for the load and hold sample. As no decrease in virus titer can be seen for the hold sample, the process conditions (e.g., temperature and buffer composition) as well as the test item itself (mAb) have no influence on the virus stability or its replication capacity. Therefore, the viral inactivation is solely caused by the low pH conditions. The columns in the middle of the diagram show virus titer and the corresponding LRV from the titration samples. The virus titers in all samples were reduced below the limit of detection, leading to the indicated minimal LRV. The slightly different minimal LRV of sampling points F and G compared with A to E is caused by different pre-dilution of these samples due to the simultaneous application of LVP assays. On the right hand side of the diagram, the LVP results from sample F and G can be seen. At this significantly higher sensitivity, no residual infectivity was found and a log reduction factor of at least 5.82 could be demonstrated.

The middle and the lower diagram of Figure 7 show the results of the two samplings with Mode B of ProtA chromatography. As described in Section 2.4.3.4, the sampling procedure was started with sampling point G for 17.31 min. After a pause of 8.66 min, the second sample for sampling point G was drawn. The further sampling was done pair-wise the same way. As already observed in Mode A, the Mode B samples were also free of virus-positive wells; no residual infectivity was found in any sample even at the higher sensitivity applied for samples F and G. As the viral inactivation is faster at lower pH levels, the results are consistent (Chinniah, Hinckley, & Connell-Crowley, 2016).

Prior to the experiment, the available literature data were analyzed to decide which sampling points should be integrated into the CFI. Based on these available data (CMC Biotech Working Group, 2009), it was expected to detect viral activity at least at the first sampling point. However, most recently published data supports the results of the discussed experiments. Chinniah et al. (2016) published viral inactivation data at different conditions over a broad range of incubation durations from 30 s up to 60 min.

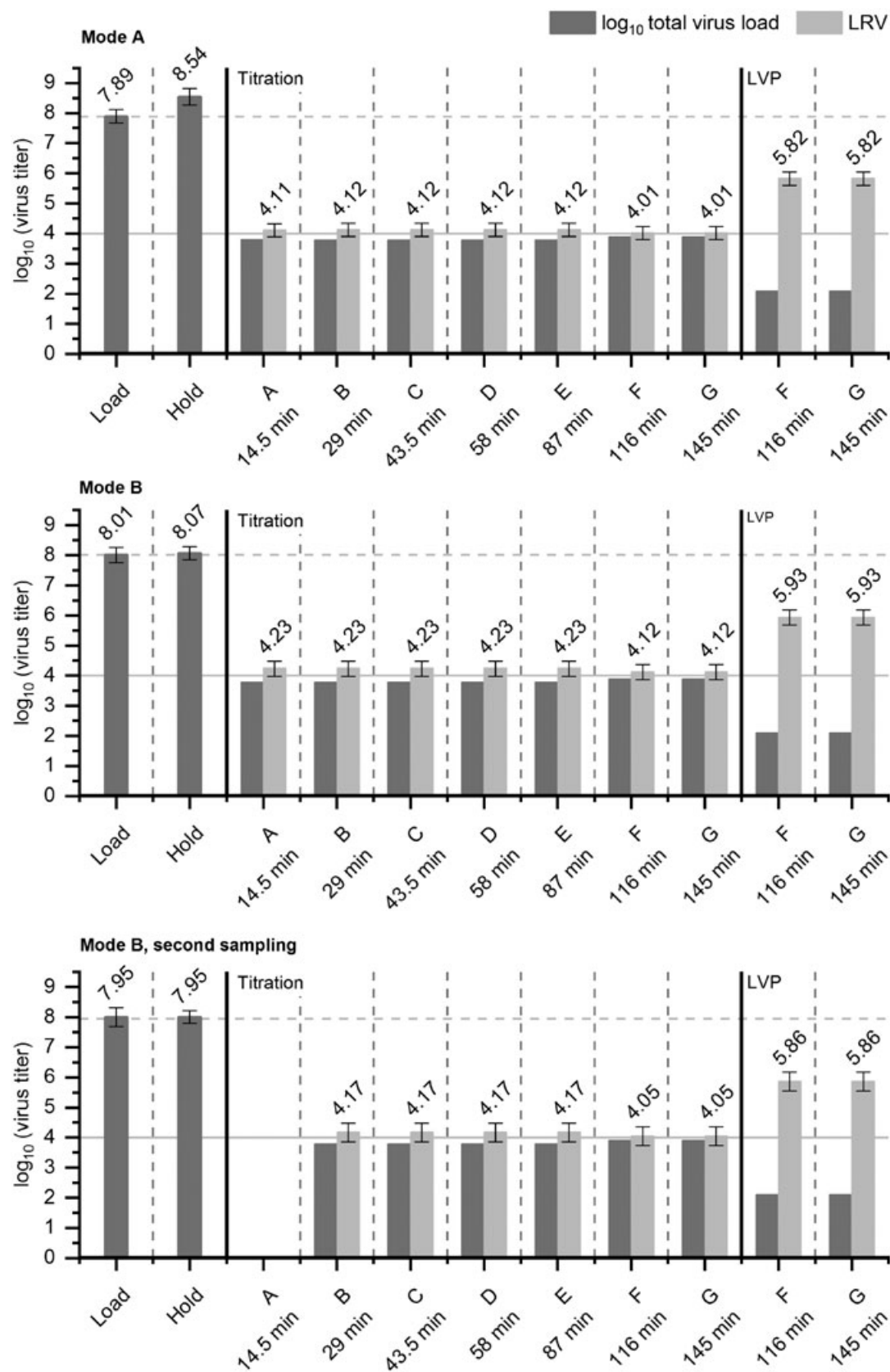


FIGURE 7 Results of virus study in a continuous process; titration and LVP results are shown. LVP: large volume plating

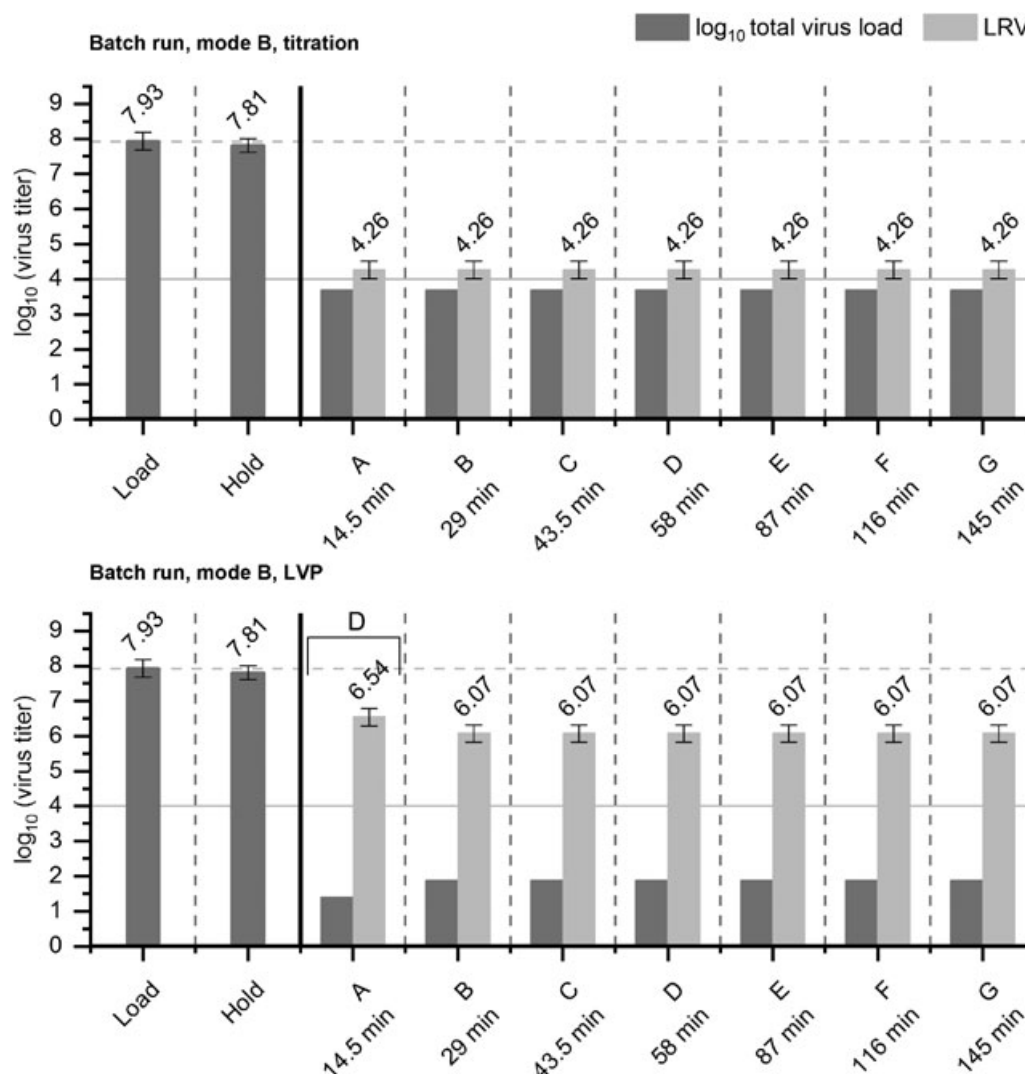


FIGURE 8 Results of virus study batch process; titration and LVP results are shown, bracket represents the sample were Mode D of LRV calculation was applied due to one virus-positive well. LRV: decadic logarithm reduction value; LVP: large volume plating

For pH values of 3.6 and 3.7, LRVs ≥ 4.7 after 3 min and ≥ 4.3 after 10 min, respectively, could be observed or interpolated. However, for pH 3.8, LRV above 4 could not be reached during the first 10 min (Chinniah et al., 2016). As shown in Table 2, both average pH levels (3.77 for Mode A, 3.44 for Mode B) lie in the range where complete viral inactivation can be expected after 14.5 min as shown in the experiment. However, the maximum pH level in the Mode A experiment with 4.07 was above the level where viral inactivation takes place ($\text{pH} < 4$). The explanation for the successful inactivation after 14.5 min despite the high maximum pH level is due to pH fluctuation with a minimum pH at 3.53. The average pH 3.77 lies within the reported pH for fast inactivation, resulting in high LRVs. In addition, computational fluid dynamics (CFD) simulation shows that the pH fluctuation is further dampened inside the CFI due to diffusion, decreasing the maximum pH over the course of the CFI (results will be published separately).

To conclude, the results are consistent with the previously published batch results.

3.4 | Virus study batch process

For the batch study, a complete eluate fraction of one column was collected for one switch time directly at the outlet of the ProtA. The average pH of this eluate fraction (84.34 ml) before the addition of spiking material was 3.55. After spiking, the samples were withdrawn at the same time points as within the CFI. All samples were analyzed with titration and LVP. The results are shown in Figure 8. The upper part represents the titration results, whereas the lower part represents the LVP results. Regarding the endpoint titration results, the viral titer in all samples was reduced below the limit of detection ($1.96 \log_{10}$ TCID₅₀/ml), leading to LRV ≥ 4.26 . In comparison with the results for the continuous process described in Section 3.3, no differences in viral inactivation capacity could be observed.

Concerning the LVP results in the lower part of Figure 8, viral activity could be detected in one sample. The LVP of Sample A (14.5 min) showed one positive well out of 384, leading to a change in LRV calculation from Mode E to Mode D and therefore to a higher

LRV. All other samples showed no viral activity. However, the higher LRV from Sample A does not hint at a better viral inactivation compared with the continuous experiments as the detection limits are not the same between the different methods. Out of the continuous runs, only the last two time points (F and G) were analyzed with LVP. Therefore, smaller sample volumes have been analyzed for the continuous experiment, leading to higher limits of detection.

To conclude, the batch experiment shows the same fast viral inactivation as the continuous experiments.

4 | CONCLUSIONS

The presented study shows the successful viral inactivation of xMuLV in continuous as well as in batch mode. Two different ProtA methods with two different pH levels showed a high inactivation capacity already during the first 14.5 min with no residual infectivity determined. The batch experiment showed similar results as the corresponding continuous experiment. The concept for a virus study for the continuous low pH viral inactivation within a CFI was developed and tested successfully.

5 | OUTLOOK

This study represents a first concept for virus studies for continuous processes and its experimental execution. The results showed clearly that the viral inactivation at the given conditions is very fast (<14.5 min). Consequently, the CFI can be shortened while still achieving reliable viral inactivation. The shortening minimizes the risk of product damage through long RT at low pH values. Moreover, it was shown that the transfer from batch results to the continuous process is possible. This is accomplished by the narrow RTD of the CFI, showing how close the setup approaches the ideal plug flow and with that batch operation. Therefore, the use of a ST would not have shown the same results as with the same reactor length, a smaller minimum RT can be reached. Future studies have to address the question, at which conditions a continuous low pH viral inactivation should be validated? Moreover, for being able to show viral inactivation kinetics inside the CFI, the prediction of experimental results needs to be improved. To enhance planning of experiments and deepen process understanding, CFD simulation should be used in the future. With the help of kinetic data from literature, a model for the simulation of viral inactivation based on varying pH inlet values should be set up. With the help of this tool, experimental results can be predicted and studies planned without the necessity of prior experiments or multiple studies.

ACKNOWLEDGMENTS

The authors would like to thank Isabell Günther, Kerstin Baumarth, Frank Seifert, and the laboratory staff of Charles River for their support before and during the study. Patent application filed by

Bayer AG, with named inventors Laura David, Benjamin Maiser, Martin Lobedann, and Peter Schwan, pending. Part of this study was funded by the German federal ministry of education and research (project WiPro, BMBF; 031A616M). Part of the equipment used in this study was previously purchased through the MoBiDiK project, co-funded by the European Union (European Regional Development Fund) and the German Federal State North Rhine-Westphalia (NRW) (NRWZiel2; 005-1010-0009).



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How to cite this article: David L, Mäiser B, Lobedann M, et al. Virus study for continuous low pH viral inactivation inside a coiled flow inverter. *Biotechnology and Bioengineering*. 2019;116: 857–869. <https://doi.org/10.1002/bit.26872>